FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Urinary Tract Infection in Pregnancy

Lecture | Week 3 · Medical disorders & PPH | Obstetrics | Dr Nanzira

Progesterone-induced ureteric relaxation and compression by the gravid uterus cause urinary stasis, so bacteriuria in pregnancy readily ascends to the kidney. Screen for and treat asymptomatic bacteriuria, treat cystitis with a pregnancy-safe antibiotic (nitrofurantoin, amoxicillin if sensitive, cephalexin) for 5–7 days, and admit every case of pyelonephritis for IV ceftriaxone, sepsis screening and fetal monitoring.

Must know before the ward round

1. Pregnancy predisposes to UTI through progesterone (smooth-muscle relaxation, ureteric dilatation, reduced peristalsis), mechanical compression (right more than left), vesicoureteric reflux, glycosuria and a larger, slower-emptying bladder.
2. Asymptomatic bacteriuria (ASB, 10^5 CFU/mL or more of one organism on a clean-catch MSU without symptoms) affects about 2–10% of pregnant women (about 13% in a Mulago study); untreated, up to 30–40% progress to pyelonephritis, so screen and treat for 7 days.
3. Dipstick nitrites are specific but only about 50% sensitive, so a negative dipstick does not exclude ASB. Send a clean-catch MSU for culture before the first antibiotic dose wherever possible.
4. Cystitis (dysuria, frequency, suprapubic pain, no fever) is treated as an outpatient for 5–7 days. Fever or loin pain means pyelonephritis, and every case is admitted.
5. E. coli causes 70–80% of UTIs; Ugandan isolates are often resistant to co-trimoxazole and amoxicillin, so culture where possible and do a test of cure.
6. First-line oral options are nitrofurantoin 100 mg MR 12-hourly, cephalexin 500 mg 8–12-hourly, or amoxicillin 500 mg 8-hourly only if sensitive, for 5–7 days, with a test-of-cure culture 1–2 weeks after treatment.
7. Safe drugs are nitrofurantoin (avoid near term and in G6PD deficiency), amoxicillin or amoxicillin-clavulanate, cephalexin, cefixime and ceftriaxone. Avoid fluoroquinolones and tetracyclines, trimethoprim in the first trimester, and sulfonamides near term.
8. Every case of pyelonephritis in pregnancy is admitted - cultures, IV ceftriaxone 1–2 g daily, careful fluids, paracetamol, fetal monitoring, then oral step-down to complete 10–14 days.
9. Screen every febrile pregnant woman for sepsis (temperature above 38 or below 36 °C, HR above 100, RR 22 or more, SBP 100 or below, confusion, SpO_2 below 95%, lactate above 2) and do an mRDT; start the sepsis bundle within 1 hour.
10. Fever persisting after 48–72 hours of IV antibiotics means check the culture (ESBL?), do a renal ultrasound for obstruction or abscess, and rethink the diagnosis.
11. Complications of pyelonephritis are sepsis/septic shock, ARDS (endotoxin capillary leak, worsened by fluid overload and beta-agonists), anaemia, AKI, preterm labour and recurrence.
12. Recurrent UTI or pyelonephritis needs nightly prophylaxis (nitrofurantoin 50–100 mg or cephalexin 250–500 mg) until after delivery, plus a search for a cause (diabetes, stones, sickle cell, HIV, anatomical anomaly).

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1. Definition

A **urinary tract infection (UTI)** in pregnancy ranges from **asymptomatic bacteriuria (ASB)**, bacteria growing silently in the bladder, through **acute cystitis** (bladder infection with symptoms) to **acute pyelonephritis** (kidney infection with fever and loin pain).

Pregnancy **does not make bacteriuria more common**, but it makes bacteriuria **far more dangerous**: the dilated, sluggish ureters let bacteria climb to the kidney. That is why a condition we usually ignore in non-pregnant women (ASB) is **screened for and treated** in pregnancy.

Term	Definition
Significant bacteriuria	**10^5 colony-forming units (CFU)/mL or more of a single uropathogen** on a clean-catch midstream urine (MSU) culture (lower counts count on a catheter specimen)
Asymptomatic bacteriuria (ASB)	Significant bacteriuria **without symptoms**
Acute cystitis (lower UTI)	Bladder infection: dysuria, frequency, urgency, suprapubic pain, haematuria; **no fever or loin pain**
Acute pyelonephritis (upper UTI)	Infection of the renal pelvis and parenchyma: **fever, rigors, loin (flank) pain, renal angle tenderness**, nausea/vomiting, ± lower urinary symptoms
Recurrent UTI in pregnancy	Two or more UTIs in pregnancy (or one UTI plus ASB), or relapse after treatment
Relapse vs reinfection	Relapse = same organism within 2 weeks (treatment failure, possible structural cause); reinfection = new organism later
Group B streptococcal (GBS) bacteriuria	GBS in urine at **any colony count**: a marker of heavy genital colonisation; treat and give **intrapartum antibiotic prophylaxis**
Urosepsis	Life-threatening organ dysfunction due to a dysregulated response to a urinary infection

> **CLINICAL PEARL**
> The one thing never to forget: **fever plus loin pain in a pregnant woman is pyelonephritis until proven otherwise, and every case is admitted.** Pyelonephritis in pregnancy is a sepsis risk, not an outpatient illness.

2. Epidemiology

- **ASB**: about **2–10%** of pregnant women worldwide. A study at **Mulago Hospital** found ASB in about **13%** of antenatal mothers, with *E. coli* the commonest isolate and high resistance to commonly used oral antibiotics Verify *Mulago study citation*.
- Untreated ASB progresses to **pyelonephritis in 20–40%** of pregnant women (versus about 1–2% if treated). Treating ASB reduces pyelonephritis by about three-quarters and is associated with fewer low-birth-weight babies.

- **Pyelonephritis** complicates about **1–2% of pregnancies**, mostly in the **second and third trimesters**, and is **more often right-sided**.
- **Why it matters in Uganda**: pyelonephritis is one of the commonest reasons for antenatal admission, a common cause of **fever in pregnancy** (alongside malaria), an important trigger of **preterm labour**, and a cause of **maternal sepsis**, a leading direct cause of maternal death. Antibiotic resistance is high, and urine culture is not available at most lower-level facilities.
- **Resistance in Uganda**: many community *E. coli* isolates are resistant to **co-trimoxazole, ampicillin/amoxicillin and tetracyclines**; **extended-spectrum beta-lactamase (ESBL)** producers are increasingly reported, including resistance to ceftriaxone. Nitrofurantoin generally retains good activity against *E. coli*.
- **Facility realities**: dipstick urinalysis is available at most HC IIIs/IVs (if strips are in stock); microscopy at HC IV; **culture and sensitivity** mainly at regional and national referral hospitals and private labs, with results in 2–3 days. **G6PD deficiency** is relatively common in Uganda (relevant to nitrofurantoin).

> **UGANDA CONTEXT**
> WHO ANC (2016) recommends **screening for ASB** in pregnancy. Where culture is not available, WHO suggests **midstream urine Gram staining** over dipstick testing, but in most Ugandan HC IIIs the practical screen is a **urine dipstick** (nitrites and leucocyte esterase) at booking and when symptomatic. Remember: **a negative dipstick does not exclude ASB** (nitrite sensitivity is only about 50%). Send urine for culture wherever you can, and always for pyelonephritis.

3. Aetiology and Risk Factors

Organisms: *Escherichia coli* (**70–80%**), *Klebsiella*, *Proteus*, *Enterobacter*, *Enterococcus*, **group B streptococcus**, *Staphylococcus saprophyticus*. Most are gut flora that colonise the perineum and ascend the short female urethra.

Risk factors

- Previous UTI or pyelonephritis.
- **Diabetes** (glycosuria).
- **Sickle cell disease and trait**.
- **HIV**.
- Urinary tract anomalies or **stones**.
- Low socio-economic status, poor hygiene/water access.
- Sexual activity, multiparity.
- **Catheterisation** (including after CS).

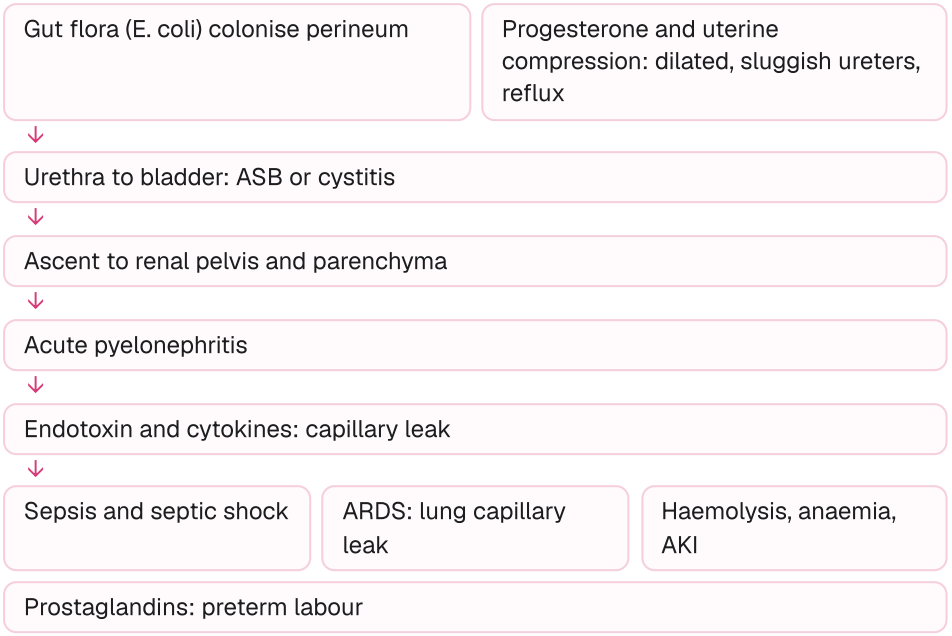
Pregnancy-specific predisposition ("PROGRESS"; mechanisms in section 4): progesterone, reflux, obstruction by the uterus, glycosuria, residual urine, *E. coli* from the nearby perineum, stasis, suppressed immunity.

4. Pathophysiology

Why pregnancy predisposes to ascending infection

Change	Mechanism	Effect
Progesterone	Relaxes smooth muscle of the ureters and bladder	**Ureteric dilatation**, reduced peristalsis, **urinary stasis**; larger, floppier bladder that empties incompletely

Change	Mechanism	Effect
Mechanical compression by the enlarging uterus (from about 20 weeks) at the pelvic brim	Ureters compressed where they cross the iliac vessels	**Physiological hydroureter and hydronephrosis** (up to 90% of women by the third trimester)
Right more than left	The uterus is **dextrorotated**; the **sigmoid colon cushions the left ureter**; the right ovarian vein crosses the right ureter	**Right-sided pyelonephritis** is more common
Vesicoureteric reflux	Bladder oedema/hyperaemia and displaced trigone reduce the valve effect at the ureteric orifices	Bacteria reflux up the ureters
Urine composition	**Glycosuria** (lower renal threshold), **aminoaciduria**, more alkaline urine	Better culture medium for bacteria
Increased GFR and urine volume	Dilated system holds a larger volume of stagnant urine (up to 200 mL)	Stasis
Immune modulation	Relative suppression of cell-mediated immunity	Infection spreads more easily
Short female urethra near the vagina and anus	Colonisation of the periurethral area by gut flora	Entry of organisms (same as non-pregnant)



Why pyelonephritis is so dangerous in pregnancy

- **Endotoxin** from Gram-negative bacteria triggers a strong cytokine response. In pregnancy, **low plasma oncotic pressure** (dilutional hypoalbuminaemia) and high plasma volume make the lungs prone to **capillary leak**: **ARDS/pulmonary oedema**, made worse by **aggressive IV fluids** and **beta-agonist tocolytics**.
- **Endotoxin damages red cell membranes** (haemolysis), causing **anaemia** in about a quarter of women.
- **Transient renal dysfunction** (fall in creatinine clearance) is common.
- **Inflammation stimulates prostaglandins and uterine contractions**: preterm labour. **Fever** itself causes fetal tachycardia and increases fetal oxygen demand.
- **ASB itself** is linked to preterm birth and low birth weight, probably through the same inflammatory pathways.

Feature links

- Bladder mucosal inflammation → dysuria (internal), frequency, urgency, suprapubic pain, haematuria → cystitis.
- Gram-negative bacteria reduce nitrate to nitrite → dipstick nitrites (specific, about 50% sensitive).
- Neutrophils in urine → leucocyte esterase, pus cells above 10/HPF; white cell casts → renal parenchymal (upper tract) infection.
- Renal parenchymal infection and capsular stretch → loin pain, renal angle tenderness; systemic cytokines → fever, rigors, vomiting.
- Dextrorotation and the sigmoid cushion → right-sided pyelonephritis.
- Endotoxin → vasodilatation and capillary leak → tachycardia, hypotension, raised lactate → sepsis/septic shock.
- Low oncotic pressure + capillary leak + IV fluids/beta-agonists → breathlessness, crackles, falling SpO_2 → ARDS/pulmonary oedema.
- Endotoxin damage to red cells → haemolysis → falling Hb (plus dilution from fluids).
- Renal inflammation and hypoperfusion → rising creatinine, oliguria → AKI.
- Prostaglandin release → contractions → preterm labour; maternal fever → fetal tachycardia.
- Obstruction (stone) or abscess → persistent fever after 48–72 hours of correct antibiotics.

5. Classification / Staging

Category	Features	Setting of care
Asymptomatic bacteriuria	Significant bacteriuria, no symptoms	Outpatient; 7 days, test of cure
Acute cystitis (lower UTI)	Dysuria, frequency, urgency, suprapubic pain; **afebrile**	Outpatient, HC III upwards; 5–7 days
Acute pyelonephritis (upper UTI)	Fever, rigors, loin pain, renal angle tenderness, vomiting	**Admit every case** (HC IV / hospital); 10–14 days in total
Urosepsis / septic shock	Pyelonephritis + sepsis triggers (or hypotension despite fluids)	Sepsis bundle within 1 hour; HDU/ICU
Recurrent UTI	Two or more UTIs, UTI + ASB, relapse; one pyelonephritis (many units)	Prophylaxis until delivery; look for a cause

Sepsis screen (suspected infection plus **any one**): temperature **above 38 °C or below 36 °C**, HR **above 100**, RR **22 or more**, SBP **100 mmHg or below**, new confusion, SpO_2 **below 95%**, oliguria, lactate **above 2 mmol/L** (severe **4 or more**).

6. Clinical Features

On an antenatal ward the question is almost always one of three: **Is this bacteriuria at all? Is it lower (cystitis) or upper (pyelonephritis)? Is she septic?** Keep an eye on the fetus and on the other causes of fever in Uganda.

History

Question	Why it matters
Burning on passing urine? Is it **internal** (in the urethra) or **external** (as urine touches sore vulval skin)?	Internal dysuria suggests UTI; external suggests vulvovaginitis (candida, trichomonas, herpes)
Frequency, urgency, suprapubic pain, blood in urine, smelly urine?	Cystitis. **Frequency alone is normal in pregnancy**: ask whether it is new or different

Question	Why it matters
Fever, rigors, loin/back pain, vomiting?	Pyelonephritis; vomiting means oral treatment will fail
Gestational age	Drug choice (no trimethoprim in the first trimester; no nitrofurantoin or sulfonamides near term), steroids if preterm labour threatens
Abdominal tightening, pain, vaginal bleeding, leaking liquor, fetal movements?	**Preterm labour**, abruption, PPROM/chorioamnionitis
Headache, myalgia, cough, breathlessness, offensive discharge, RIF pain?	Other fever sources: malaria, pneumonia, chorioamnionitis, appendicitis; **breathlessness during treatment suggests ARDS/fluid overload**
Vaginal discharge, itching, new partner?	STI (chlamydia/gonorrhoea cause dysuria with sterile pyuria), vaginitis
Previous UTIs this pregnancy: treated with what, and was a **test of cure** done? Previous pyelonephritis, kidney stones, known urinary tract abnormality?	Recurrence or relapse changes the plan (culture, prophylaxis, imaging)
Diabetes, **sickle cell disease/trait**, **HIV** (and co-trimoxazole use), TB, recent catheter	Risk factors; atypical organisms; drug choices
Antibiotics already taken (including pharmacy-bought ciprofloxacin)	Affects culture yield and resistance
Drug allergies (penicillin: type of reaction), **G6PD deficiency**	Choice of antibiotic (nitrofurantoin; cephalosporins after anaphylaxis)
Malaria test done? Travel, ITN use	**In Uganda every febrile pregnant woman needs an mRDT/blood smear**

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), with the obstetric abdominal examination. A febrile, unwell woman gets her vital signs and sepsis screen first (section 9).

Introduction, consent and preparation

- Introduce yourself, confirm identity and gestation, explain and gain consent; chaperone for abdominal, speculum and vaginal examination.
- **Hand hygiene**.
- **Position and exposure**: semi-recumbent with **left lateral tilt**; sit her forward for the renal angles.

General inspection and vital signs

- **Vital signs are the priority**: temperature, pulse, **respiratory rate**, BP, SpO_2, level of consciousness, urine output, on a **modified obstetric early warning score (MEOWS)** chart. Tachycardia above 100, RR 22 or more, SBP 100 or below, or confusion suggest **sepsis**.
- **General**: toxic look, rigors, hydration, pallor (haemolysis), jaundice, rash, confusion.

Hands, face and neck

Warm flushed peripheries (early sepsis) versus cold mottled skin and slow capillary refill (shock); conjunctival pallor; scleral jaundice; dry mucous membranes.

Chest

Crackles (pneumonia; later, pulmonary oedema/ARDS). **Re-examine after every fluid bolus.**

Abdomen and obstetric examination

- **Inspection**: size for dates, scars.

- **Palpation**: suprapubic tenderness (cystitis); bladder (retention); right iliac fossa/flank tenderness (appendicitis: the appendix moves upwards and laterally in pregnancy); RUQ (cholecystitis, HELLP); **uterine tenderness** (chorioamnionitis, abruption, red degeneration of a fibroid); **contractions**.
- **Renal angle (costovertebral angle) tenderness**: sit her forward and percuss gently (pyelonephritis).
- **Fundal height, lie, presentation, engagement**.
- **Auscultation of the fetal heart**: **fetal tachycardia** with maternal fever.

Completing the examination

- **Speculum** if discharge, bleeding or leaking (PPROM, cervicitis).
- **Vaginal examination** only if contractions or suspected labour (and not if PPROM without labour): cervical change = preterm labour.
- **Bedside tests**: urine dipstick, mRDT, Hb (section 8).
- Thank her, hand hygiene, summarise.

Document key positives and negatives

Area	Model documentation
General	"Febrile, flushed, dry mucosae, not pale, alert"
Vital signs	"T 39.2 °C, HR 118, RR 22, BP 104/66, SpO_2 97% on air: sepsis screen positive"
Chest	"Chest clear" (re-examine after fluids)
Abdomen	"Right renal angle tenderness; no RIF tenderness; bladder not palpable"
Obstetric	"Uterus soft, non-tender, no contractions, FHR 170"
Speculum	"No liquor, no discharge"

Red flags

RED FLAG

- **Signs of sepsis**: temperature above 38 °C or below 36 °C, HR above 100 (persistently above 110), RR 22 or more, SBP 100 mmHg or below, confusion, SpO_2 below 95%, oliguria (below 0.5 mL/kg/h), mottled skin, raised lactate. Start the **sepsis bundle within 1 hour**.
- **Breathlessness, falling SpO_2 or crackles** during treatment (especially after IV fluids): **ARDS or pulmonary oedema** until proven otherwise; stop IV fluids, sit her up, oxygen, senior review.
- **Persistent fever after 48–72 hours** of appropriate IV antibiotics: resistant organism, **obstruction (stone), renal or perinephric abscess**, or wrong diagnosis.
- **Regular contractions or cervical change**: preterm labour.
- **Reduced fetal movements, fetal tachycardia above 160 or bradycardia**.
- **Rising creatinine / oliguria**: AKI.
- **Vomiting**: cannot take oral treatment; needs IV therapy.

Thinking like a clinician

- **In Uganda, fever in pregnancy is malaria AND something else until proven otherwise.** Always do an mRDT, but a positive mRDT does not exclude pyelonephritis (both can co-exist; the mRDT can stay positive for weeks after treated malaria).
- **Distinguish cystitis from pyelonephritis clinically**: **fever and loin pain = upper tract = admit**. Cystitis does not cause high fever.
- **Pyuria is not bacteriuria.** Leucocytes without nitrites and a negative culture (sterile pyuria) suggest **chlamydia/gonorrhoea, TB of the urinary tract, contamination with vaginal discharge, or recent**

antibiotics.

- **Contaminated samples** are common: many squamous epithelial cells and mixed growth mean the MSU was not clean. Repeat it.
- **Right loin pain has a big differential** in pregnancy: pyelonephritis, appendicitis, renal colic, cholecystitis, HELLP, abruption, red degeneration of a fibroid, ovarian torsion. Examine the uterus and check urine, BP and bloods.
- **When the patient does not improve**, do not simply add more antibiotics: check the culture result, image the kidneys (ultrasound for obstruction/abscess), reconsider the diagnosis (malaria, chorioamnionitis, appendicitis, pneumonia).
- **Over-hydration is a real danger**: pregnant women with pyelonephritis are prone to pulmonary oedema. Give fluids in small boluses with reassessment and keep a fluid balance chart.
- **Common student mistakes**:
  1. Prescribing **ciprofloxacin** (commonly used for UTI in non-pregnant patients in Uganda) to a pregnant woman.
  2. Treating ASB with a **single dose** or 3 days by default (use 5–7 days; WHO recommends 7 days for ASB).
  3. Forgetting a **test of cure** urine culture 1–2 weeks after treatment.
  4. Sending a woman with pyelonephritis home on oral antibiotics.
  5. Using **nitrofurantoin for pyelonephritis** (it does not achieve adequate renal tissue or blood levels).
  6. Forgetting the **fetus** (fetal heart, contractions, steroids if preterm).

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Acute cystitis	Dysuria, frequency, urgency, suprapubic pain; afebrile	Dipstick (nitrites, LE), MSU culture
Acute pyelonephritis	Fever, rigors, loin pain, renal angle tenderness, vomiting	Urinalysis, culture, FBC, blood culture, creatinine
Malaria	Fever, rigors, headache, myalgia; no urinary symptoms (but can co-exist)	mRDT / blood smear
Vulvovaginal candidiasis / trichomoniasis	External dysuria, itch, discharge	Speculum, wet mount
Urethritis / cervicitis (chlamydia, gonorrhoea)	Dysuria, discharge, sterile pyuria	NAAT where available, syndromic management
Preterm labour	Regular painful contractions, cervical change; may be triggered by UTI	Speculum/VE, CTG, urine culture
Chorioamnionitis	Fever, **uterine tenderness**, offensive liquor, fetal tachycardia, often after PROM	Clinical, FBC, HVS
Placental abruption	Constant pain, tender hard uterus, bleeding, fetal distress	Clinical, CTG, ultrasound
Appendicitis	Right-sided pain (higher in pregnancy), anorexia, low-grade fever, guarding; urinary symptoms can occur if inflamed appendix lies near the ureter	Clinical, ultrasound
Renal colic / urolithiasis	Severe colicky loin-to-groin pain, haematuria, little fever unless infected obstruction	Renal ultrasound (distinguish from physiological hydronephrosis)

Differential	Distinguishing clinical feature	Distinguishing investigation
Pre-eclampsia / HELLP	RUQ/epigastric pain, hypertension, proteinuria	BP, urine protein, FBC (platelets), LFTs
Red degeneration of fibroid	Localised tender mass on the uterus, mid-pregnancy, low-grade fever	Ultrasound
Pneumonia (lower lobe)	Cough, pleuritic pain, crackles; pain can refer to the loin	CXR (shielded)
Urinary TB	Chronic symptoms, sterile pyuria, HIV-positive, weight loss	Early-morning urine for GeneXpert/AFB

8. Clinical Workup

Availability reflects the source notes Verify *compiled from the source notes, not an MoH list*.

Bedside

Test	How to read it	Available at
Urine dipstick (fresh sample)	**Nitrites positive**: Gram-negative bacteria (specific; only about 50% sensitive). **Leucocyte esterase**: pyuria. Both positive with symptoms = treat as UTI. LE alone = possible contamination, STI or vaginitis. **Protein**: UTI or pre-eclampsia (check BP). **Glucose**: lower renal threshold or diabetes. **Ketones**: vomiting/starvation. **Blood**: infection, stone, contamination	Most HC IIIs and IVs (if strips in stock)
Look at the urine	Cloudy, offensive urine supports infection; clear urine makes it less likely (not excluded)	All levels
mRDT (± blood smear)	Every febrile woman. A positive HRP2 mRDT can stay positive for weeks after treated malaria and does **not** exclude pyelonephritis	All levels
RBS	Undiagnosed diabetes (recurrent UTI) or stress hyperglycaemia	HC III upwards
HemoCue Hb	Haemolytic anaemia from endotoxin	HC III upwards
HIV rapid test (if status unknown)	Risk factor; changes prophylaxis (co-trimoxazole)	All levels
Pulse oximetry	Baseline and after every fluid bolus (ARDS watch)	HC IV upwards
Hourly urine output (catheter if septic)	Aim **0.5 mL/kg/h or more**	All inpatient levels
Fetal heart / CTG (viable gestation)	Tachycardia settles as maternal fever falls; persistent abnormality needs review	All levels (CTG at hospitals)
Point-of-care ultrasound	Fetal viability, liquor, placenta; **gross hydronephrosis** if trained (formal renal scan still needed)	HC IV and hospitals

How to collect a clean-catch MSU (teach the patient). The sample itself is where most mistakes happen.

1. Wash hands; separate the labia with one hand.
2. Clean the vulva **front to back** with water (not antiseptic, which can kill bacteria in the sample).
3. Pass the **first part of the stream into the toilet**, then catch the **middle part** in a sterile container without touching the inside.
4. Deliver to the lab **within 2 hours** (or refrigerate at 4 °C for up to 24 hours).
5. Label the form with **gestation, symptoms and any antibiotics already taken**, and whether it is a catheter specimen (lower counts are significant).

Basic lab

Test	When	Expected / abnormal finding	How it changes management (and where)
Urine microscopy	Symptoms, positive dipstick	Pus cells **more than 10 per high-power field**, bacteria; **white cell casts = pyelonephritis**; many squamous cells = contaminated sample	Supports diagnosis; repeat a contaminated sample. HC IV and above
Gram stain of MSU	Where culture is unavailable (WHO-suggested ASB screen)	Bacteria on Gram stain	Treat ASB empirically, then test of cure if possible
FBC	Pyelonephritis, sepsis	WBC raised; **anaemia** (haemolysis, about a quarter); **low platelets** (sepsis/DIC)	Severity; transfusion; DIC workup. HC IV upwards
Creatinine, urea, electrolytes	Pyelonephritis, sepsis, before gentamicin	Creatinine above about 77–80 µmol/L is suspicious in pregnancy (normal 35–70)	AKI: adjust or avoid gentamicin; fluid balance. Hospitals
LFTs	RUQ pain, jaundice, hypertension	Raised ALT, bilirubin	HELLP, cholecystitis, hepatitis. Hospitals
Lactate (where available)	Sepsis screen positive	**Above 2 mmol/L** = hypoperfusion; **4 or more** = severe	Fluid boluses, HDU/ICU; lactate above 2 alone meets the sepsis trigger

Imaging

Test	When	Expected finding	How it changes management (and where)
Renal ultrasound	**No response in 48–72 h**, recurrent pyelonephritis, haematuria or colic, suspected stone	**Hydronephrosis out of proportion** to gestation, stone, renal/perinephric abscess	Urology: ureteric stent or nephrostomy; drainage. Hospitals
Obstetric ultrasound	Pyelonephritis, contractions, uncertain dates	GA, liquor, placenta, fetal wellbeing	Steroids, timing, differential (abruption, fibroid degeneration). HC IV and hospitals
Chest X-ray (shielded)	Breathless, SpO_2 below 95%, crackles	Bilateral infiltrates (ARDS/oedema), lobar consolidation	Stop fluids, oxygen, diuretic if oedema, ICU. Hospitals
Renal imaging postpartum	Recurrent pyelonephritis or suspected structural disease	Anomaly, scarring, stones	After **6–12 weeks postpartum**, when physiological dilatation has resolved

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management (and where)
Urine culture and sensitivity (MSU before antibiotics)	**Booking screen** (ASB), every symptomatic UTI where possible, **always in pyelonephritis**, recurrence, treatment failure	**10^5 CFU/mL or more of one organism** (lower counts on a catheter specimen); **GBS at any count**; mixed growth = contamination	Targets therapy (resistance to amoxicillin/co-trimoxazole, **ESBL**); GBS → intrapartum prophylaxis. Regional/national referral and private labs; results in 2–3 days
Test-of-cure culture	**1–2 weeks** after finishing treatment	Persistent bacteriuria in about 1 in 5	Re-treat by sensitivity; consider prophylaxis
Blood cultures	Pyelonephritis, sepsis, before antibiotics	Positive in **15–20%** of pyelonephritis	Guides antibiotics; bacteraemia = closer monitoring. Regional/national referral
Early-morning urine for GeneXpert/AFB	Sterile pyuria, HIV, weight loss	MTB detected	TB treatment
NAAT for chlamydia/gonorrhoea	Sterile pyuria, discharge	Positive	Treat woman and partner; otherwise syndromic management

Putting it together

Model one-line summary

> "A 24-year-old G2P1 at 26 weeks with 2 days of fever, rigors, right loin pain and vomiting, febrile 39.2 °C, tachycardic 118, RR 22, BP 104/66, with right renal angle tenderness, nitrite-positive urine, negative mRDT and fetal tachycardia."

Problem list

1. **Acute right pyelonephritis with sepsis** (sepsis screen positive).
2. Vomiting and dehydration: needs IV therapy, but at risk of fluid overload/ARDS.
3. **Fetal tachycardia** and risk of **preterm labour** at 26 weeks.
4. Previous untreated cystitis at 14 weeks without test of cure: recurrence risk, needs prophylaxis.

Working diagnosis: acute pyelonephritis in pregnancy complicated by sepsis, probably *E. coli*.

Model case presentation (ward round)

> "Mrs B is a 24-year-old G2P1 at **26 weeks**, admitted last night with **2 days of fever, rigors, right loin pain and vomiting**, with a history of cystitis at 14 weeks that was not followed up. On admission: T 39.2 °C, HR 118, RR 22, BP 104/66, SpO_2 97%, **right renal angle tenderness**, uterus soft and non-tender, no contractions, FHR 170. Dipstick: nitrites positive, LE 3+. **mRDT negative.** Hb 10.1 g/dL, WBC 17, creatinine 88 µmol/L. Impression: **acute right pyelonephritis with sepsis** (MEOWS triggers). She received **ceftriaxone 2 g IV** within an hour after blood and urine cultures, 1 L Ringer's lactate in boluses, and paracetamol. This morning: T 37.8 °C, HR 96, RR 18, urine output adequate, FHR 150, no contractions. **Plan**: continue ceftriaxone until afebrile 24–48 hours, then **oral cefixime to complete 14 days** guided by culture; fluid balance; watch SpO_2 for ARDS; CTG/contractions; **test of cure** in 1–2 weeks, then **nightly nitrofurantoin prophylaxis** and monthly urine cultures."

Worked example 1: reading urine results

Woman	Symptoms	Dipstick	Microscopy / culture	Interpretation and action
A (booking, 14 weeks)	None	Nitrites **negative**, LE negative	Culture: **10^5 CFU/mL *E. coli***, sensitive to nitrofurantoin and cephalexin	**ASB** (a negative dipstick did not exclude it). Nitrofurantoin 100 mg MR 12-hourly for **7 days**; test of cure in 1–2 weeks
B (22 weeks)	Dysuria, frequency, no fever	Nitrites **+**, LE **++**	Pus cells 25/HPF; culture pending	**Acute cystitis**. Send MSU first, start **cephalexin 500 mg 8-hourly for 5–7 days**; adjust to sensitivity; safety-net advice
C (20 weeks)	Dysuria, itch, discharge	Nitrites negative, LE **+**	Pus cells 15/HPF, many squamous cells; culture: **mixed growth**	**Contaminated sample** and likely **vulvovaginitis**. Speculum, treat the discharge, repeat a proper clean-catch MSU
D (30 weeks)	None	Negative	Culture: **GBS 10^4 CFU/mL**	**GBS bacteriuria** (any count counts): treat now and give **intrapartum IV benzylpenicillin** in labour
E (18 weeks)	Dysuria, weight loss, HIV-positive	LE **++**, nitrites negative	Pus cells 30/HPF, **culture sterile** twice	**Sterile pyuria**: think chlamydia/gonorrhoea, recent antibiotics, **urinary TB** (early-morning urine for GeneXpert)

Worked example 2: the sepsis screen in pyelonephritis (admission observations for Mrs B)

Parameter	Value	Trigger?
Temperature	39.2 °C	Yes (above 38)
Heart rate	118/min	Yes (above 100)
Respiratory rate	22/min	Yes (22 or more)
Systolic BP	104 mmHg	No (but close to 100)
SpO_2	97%	No
Consciousness	Alert	No
Lactate	2.6 mmol/L	Yes (above 2)
WBC / Hb / creatinine	17×10^9/L / 10.1 g/dL / 88 µmol/L	Leucocytosis; creatinine **raised for pregnancy** (AKI risk)

Interpretation: sepsis from a urinary source (several triggers plus raised lactate), not yet septic shock.

Action within 1 hour: senior informed; oxygen only if SpO_2 falls below 94%; **blood and urine cultures**, then **ceftriaxone 2 g IV** (avoid gentamicin for now given the creatinine); **500 mL Ringer's lactate** boluses with chest and SpO_2 checks after each; catheter and hourly urine output; paracetamol 1 g; repeat lactate. FBC and creatinine daily. CTG: expect the fetal tachycardia to settle as the fever falls. **Review at 48–72 hours**: if still febrile, check culture (ESBL?) and arrange a **renal ultrasound**.

9. Management

Immediate / emergency

First 5 minutes (febrile pregnant woman)

1. **Full set of vital signs on a MEOWS chart**: temperature, pulse, **RR**, BP, SpO_2, level of consciousness, and the time she last passed urine.
2. **Sepsis screen** (section 5). If positive: **call a senior** and start the **sepsis bundle within 1 hour**.
3. **IV access and bloods at the same time**: FBC, **blood cultures**, creatinine, **mRDT**, glucose, lactate where available, group and save if anaemic.

4. **Urine sample before the first antibiotic dose**: clean-catch MSU (or catheter specimen if she is too ill), for dipstick and **culture**.
5. **Start treatment without waiting for results**: **ceftriaxone 1 g IV daily (2 g if septic)**, paracetamol 1 g, and **500 mL crystalloid boluses** only if hypotensive, reassessing the chest and SpO_2 after each.
6. **Fetal heart and uterus**: FHR (tachycardia above 160 is common with maternal fever), contractions, uterine tenderness.

> **RED FLAG**
> **Breathless, SpO_2 falling or crackles after IV fluids** = pulmonary oedema/ARDS until proven otherwise. Stop the fluids, sit her up, give oxygen and call for help.

Urosepsis and septic shock (algorithm)

1. **Recognise**: suspected infection + abnormal MEOWS (e.g. SBP 100 or below, RR 22 or more, altered mentation) or lactate above 2.
2. **Call for help** (senior obstetrician, anaesthetist/ICU).
3. **Within the first hour ("Sepsis Six")**:
    1. **Oxygen** to keep SpO_2 94–98%.
    2. **Blood cultures** (and urine culture).
    3. **IV broad-spectrum antibiotics**: **ceftriaxone 2 g IV** (± gentamicin 5 mg/kg; meropenem if known ESBL or hospital-acquired).
    4. **IV fluids**: 500 mL crystalloid boluses, reassessing after each (up to about 30 mL/kg).
    5. **Lactate and Hb**.
    6. **Urine output**: catheterise, hourly measurement.
4. **Left lateral tilt** (relieve aortocaval compression) if more than 20 weeks.
5. **Persistent hypotension after fluids**: vasopressors (noradrenaline) in HDU/ICU.
6. **Source control**: relieve obstruction (stent/nephrostomy); drain abscess.
7. **Fetus**: CTG once the mother is stable; the fetal heart usually recovers with maternal resuscitation. **Deliver only for obstetric indications** or if intrauterine infection is the source.
8. **VTE prophylaxis**, glucose control, reassess every 30–60 minutes.

Definitive

1. Screening and treating asymptomatic bacteriuria

1. **Screen at booking** (ideally 12–16 weeks): urine culture where available; otherwise Gram stain or dipstick (and culture if dipstick positive).
2. **If significant bacteriuria**: treat according to sensitivities for **5–7 days** (WHO: a **7-day** regimen for ASB).
3. **Test of cure**: repeat culture **1–2 weeks after** completing treatment, then **each trimester** (or monthly in high-risk women).
4. **Two episodes** (or one episode that relapses): consider **suppressive prophylaxis** for the rest of pregnancy.
5. **GBS in urine (any count)**: treat now **and** give **intrapartum IV antibiotic prophylaxis** in labour (benzylpenicillin 3 g IV loading, then 1.8 g IV 4-hourly until delivery; clindamycin 900 mg IV 8-hourly if allergic).

2. Acute cystitis (outpatient, HC III upwards)

1. Confirm: symptoms + dipstick (nitrites/LE); **send MSU for culture before the first dose** where possible.
2. Start **empirical oral antibiotic** (drug table), adjust to sensitivities when available.

3. **Advice**: drink plenty of fluids (2–3 L/day), void regularly and after intercourse, wipe front to back, paracetamol for discomfort.
4. **Return immediately** if fever, loin pain, vomiting, contractions or bleeding.
5. **Test of cure** 1–2 weeks after treatment.

3. Acute pyelonephritis: admit every case (HC IV and hospital)

1. **Assess ABCDE and screen for sepsis** (MEOWS; temperature, HR, RR, BP, SpO_2, mental state, urine output).
2. **IV access**; bloods: **FBC, creatinine and electrolytes, blood cultures, mRDT/smear, glucose, lactate** (where available), group and save if anaemic.
3. **Urine for culture before antibiotics** (MSU or catheter specimen).
4. **Start IV antibiotics immediately** (within 1 hour if septic): **ceftriaxone** first line; ampicillin + gentamicin where ceftriaxone is unavailable; meropenem for ESBL or ceftriaxone failure (doses in the drug table). Continue IV until **afebrile for 24–48 hours**, then oral step-down to complete **10–14 days**.
5. **Fluids**: IV crystalloid (Ringer's lactate or 0.9% saline). If hypotensive: **500 mL bolus**, reassess, repeat to a total of about 30 mL/kg while watching for crackles and falling SpO_2. Otherwise maintenance fluids while vomiting. **Strict fluid balance chart**; aim for urine output **0.5 mL/kg/hour or more**. Stop fluids if she becomes breathless.
6. **Antipyretic and analgesia**: **paracetamol 1 g orally/IV 6-hourly** (max 4 g/day); fever increases fetal oxygen demand and uterine irritability. Avoid NSAIDs after 28 weeks.
7. **Antiemetic** if vomiting (e.g. metoclopramide 10 mg IV/IM 8-hourly).
8. **Treat co-existing malaria** if the test is positive (IV artesunate for severe malaria).
9. **Fetal assessment**: fetal heart rate (tachycardia usually settles with maternal fever), CTG if viable gestation; watch for contractions.
10. **Preterm labour**: give **corticosteroids** if 24–34 weeks and delivery is likely (dexamethasone 6 mg IM 12-hourly × 4). Contractions often settle once fever is controlled. **Avoid beta-agonist tocolytics** (salbutamol) because they increase the risk of **pulmonary oedema/ARDS**; if tocolysis is appropriate, use **nifedipine** and only when infection is being controlled.
11. **Thromboprophylaxis** (LMWH) if immobile, septic or with other risk factors.
12. **Monitor**: vital signs 1–4-hourly (MEOWS), fluid balance, SpO_2, daily FBC and creatinine in severe cases.
13. **Review at 48–72 hours**:
    - **Improving**: switch to oral when afebrile 24–48 hours and eating; discharge with the oral course.
    - **Not improving**: check culture/sensitivity, change antibiotics, **renal ultrasound** (obstruction, stone, abscess), reconsider diagnosis, involve urology (ureteric stent or percutaneous nephrostomy for infected obstruction).
14. **After treatment**: **test of cure** 1–2 weeks after, then **suppressive prophylaxis** for the rest of pregnancy (recurrence about 6–8%, higher without prophylaxis) and **monthly or each-trimester urine culture**.

4. Recurrent UTI and prophylaxis

Indications: two or more UTIs in pregnancy, one UTI plus ASB, one episode of pyelonephritis (many units), or persistent bacteriuria after two courses.

- **Prophylaxis**: nitrofurantoin 50–100 mg nightly (or after intercourse) until delivery (switch to cephalexin from about 36 weeks), or cephalexin 250–500 mg nightly until delivery or up to 6 weeks postpartum.
- **Look for a cause**: diabetes (glucose), **sickle cell disease/trait**, HIV, **renal ultrasound** for stones/anomalies (and full imaging postpartum if recurrent), incomplete emptying.
- **Monthly urine culture** for the rest of the pregnancy.
- Behavioural advice: fluids, regular and post-coital voiding, front-to-back wiping.

5. Postpartum

- **Postpartum UTI** is common after catheterisation and CS; treat as above (nitrofurantoin is compatible with breastfeeding except for G6PD-deficient or jaundiced/preterm infants).
- Women with **recurrent pyelonephritis** or suspected structural disease need **renal imaging after 6–12 weeks postpartum** (when physiological dilatation has resolved).

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Nitrofurantoin	**100 mg** (modified-release) or 50–100 mg (standard)	Oral, with food	**12-hourly** (MR) or 6-hourly (standard)	**5–7 days** (ASB: 7 days)	**Not for pyelonephritis**; **avoid at term (from about 36–38 weeks), in labour, and in maternal G6PD deficiency** (haemolysis in mother or neonate); avoid if eGFR below 45
Amoxicillin	**500 mg**	Oral	**8-hourly**	**7 days**	**Only if the organism is sensitive** (high resistance in Uganda)
Amoxicillin-clavulanate	**625 mg** (500/125)	Oral	**8-hourly**	**5–7 days** (step-down to complete 10–14 days)	Broader cover; useful empirical choice
Cephalexin (cefalexin)	**500 mg**	Oral	**8–12-hourly** (up to 6-hourly)	**5–7 days**	Safe throughout pregnancy; avoid after anaphylaxis to penicillin
Cefixime	**200 mg** (or 400 mg once daily)	Oral	**12-hourly**	**5–7 days**; step-down to complete 10–14 days	Oral third-generation cephalosporin; good step-down option
Fosfomycin (where available)	**3 g**	Oral	Single dose	Once	Safe; limited availability in Uganda
Ceftriaxone (pyelonephritis first line)	**1 g** (2 g if septic or severe)	IV (or IM)	**Once daily**	Until **afebrile for 24–48 hours**, then oral step-down (total 10–14 days)	ESBL resistance increasing
Ampicillin + gentamicin (alternative)	Ampicillin **2 g** + gentamicin **5 mg/kg**	IV	Ampicillin 6-hourly; gentamicin once daily	Short course	Check renal function; fetal oto/nephrotoxicity with prolonged use
Meropenem (ESBL or ceftriaxone failure, referral hospital)	**1 g**	IV	**8-hourly**	Per culture	Reserve drug
Paracetamol	**1 g**	Oral or IV	6-hourly	While febrile	Max 4 g/day; avoid NSAIDs after 28 weeks
Metoclopramide	**10 mg**	IV or IM	8-hourly	While vomiting	

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Ringer's lactate / 0.9% saline	**500 mL** boluses if hypotensive (up to about 30 mL/kg); otherwise maintenance	IV	Reassess after each bolus		ARDS/pulmonary oedema: stop if breathless
Dexamethasone (preterm labour 24–34 weeks)	**6 mg**	IM	12-hourly	4 doses	
Nifedipine (tocolysis, if appropriate)	Per unit protocol	Oral			Only when infection is being controlled; **no salbutamol**
Nitrofurantoin prophylaxis	**50–100 mg**	Oral	**Once nightly** (or after intercourse)	Until delivery (switch to cephalexin from about 36 weeks)	As above
Cephalexin prophylaxis	**250–500 mg**	Oral	**Once nightly**	Until delivery or up to 6 weeks postpartum	
Benzylpenicillin (GBS intrapartum prophylaxis)	**3 g** loading, then **1.8 g**	IV	Then 4-hourly	Until delivery	Allergy: **clindamycin 900 mg IV 8-hourly**

Penicillin allergy: nitrofurantoin or (if mild non-anaphylactic allergy) cephalexin; avoid cephalosporins after anaphylaxis to penicillin.

DRUG DOSING

Antibiotics to avoid in pregnancy

- **Fluoroquinolones** (ciprofloxacin, levofloxacin): cartilage/joint damage in animal studies; avoid (use only if no alternative, on specialist advice).
- **Tetracyclines** (doxycycline): **tooth discolouration and impaired fetal bone growth**; maternal hepatotoxicity.
- **Trimethoprim**: a **folate antagonist**; avoid in the **first trimester** (neural tube and other defects).
- **Sulfonamides / co-trimoxazole** (as UTI treatment): avoid in the **first trimester** and **near term** (displaces bilirubin from albumin: **neonatal jaundice/kernicterus**; haemolysis in G6PD deficiency). HIV-positive women on **co-trimoxazole prophylaxis** continue it (the benefit outweighs the risk), but it should not be relied on as UTI treatment.
- **Aminoglycosides** (gentamicin): **fetal ototoxicity/nephrotoxicity** risk; use only short courses when clearly needed (e.g. severe sepsis, resistant organisms), with monitoring.
- **Nitrofurantoin near term**: neonatal haemolysis (especially in G6PD deficiency).

Referral criteria

Level	What to do	Refer up when
HC III	ASB and cystitis: dipstick, oral antibiotics, advice, test of cure where culture can be sent. Pyelonephritis: recognise, give **the first dose of ceftriaxone 1 g IM/IV** (and antimalarial if mRDT positive), paracetamol, oral fluids	**Every pyelonephritis** → HC IV / hospital
HC IV / General Hospital	Admit pyelonephritis; IV ceftriaxone, fluids, monitoring, microscopy, obstetric ultrasound	Septic shock, respiratory distress, AKI, no improvement in 48–72 hours, preterm labour below 34 weeks without neonatal care, or need for imaging/culture/urology → RRH
RRH	Culture and sensitivity, blood cultures, renal ultrasound, HDU, meropenem, urology, neonatal unit	ICU/ventilation, nephrostomy or complex urology unavailable → NRH
NRH	ICU, interventional urology	n/a

Monitoring and follow-up

- **Pyelonephritis**: vital signs 1–4-hourly (MEOWS); **SpO_2 and chest after every bolus**; strict fluid balance, urine output 0.5 mL/kg/h or more; daily FBC and creatinine in severe cases; fetal heart/CTG and contractions; **review at 48–72 hours**.
- **Test of cure**: urine culture **1–2 weeks** after every treated episode.
- **Ongoing**: urine culture each trimester (monthly if high-risk or after pyelonephritis); prophylaxis until delivery where indicated.
- **Postpartum**: renal imaging at 6–12 weeks if recurrent pyelonephritis or suspected structural disease.

10. Complications

Disease-related

Maternal	Fetal / neonatal
Pyelonephritis (from untreated ASB/cystitis)	**Preterm birth**
Sepsis and septic shock	**Low birth weight**
ARDS / pulmonary oedema (1–2% of pyelonephritis)	Fetal tachycardia, hypoxia, **stillbirth** in septic shock
Anaemia (haemolysis)	Neonatal sepsis (e.g. GBS)
Acute kidney injury (usually transient)	Association with hypertensive disorders and growth restriction (ASB)
Perinephric/renal abscess, **infected obstruction**	
Recurrence (6–8%, higher without prophylaxis)	
Chronic pyelonephritis/renal scarring (rare, recurrent infection with reflux)	
Maternal death (sepsis)	

Treatment-related

Complication	Cause	Recognition and action
Pulmonary oedema / ARDS	Aggressive IV fluids; beta-agonist tocolytics	500 mL boluses with reassessment; no salbutamol

Complication	Cause	Recognition and action
Neonatal haemolysis	Nitrofurantoin near term or in G6PD deficiency	Avoid from about 36–38 weeks; cephalexin instead
Neonatal jaundice / kernicterus	Sulfonamides near term	Avoid
Fetal oto/nephrotoxicity	Prolonged aminoglycosides	Short courses; check renal function
First-trimester defects	Trimethoprim (folate antagonist)	Avoid in the first trimester
Tooth discolouration, bone growth	Tetracyclines	Avoid
Treatment failure	Resistance (ESBL, amoxicillin, co-trimoxazole)	Culture before antibiotics; test of cure

11. Prognosis

- **Treated ASB** progresses to pyelonephritis in only about 1–2% (versus 20–40% untreated); treatment reduces pyelonephritis by about three-quarters and is associated with fewer low-birth-weight babies.
- **Pyelonephritis** usually responds within 48–72 hours of appropriate IV antibiotics; failure to respond suggests resistance, obstruction, abscess or a wrong diagnosis.
- **Recurrence** after pyelonephritis is about **6–8%**, higher without prophylaxis.
- Renal dysfunction is usually transient; chronic scarring is rare.

12. Prevention

- **Screen for ASB at booking** (ideally 12–16 weeks) and treat; **test of cure**; repeat culture each trimester (monthly if high-risk).
- **Treat cystitis promptly** with a pregnancy-safe antibiotic and do a test of cure.
- **Prophylaxis** after recurrence or pyelonephritis until delivery.
- **Behavioural advice**: fluids (2–3 L/day), regular and post-coital voiding, front-to-back wiping.
- **Avoid unnecessary catheterisation**; remove catheters early.
- **Antimicrobial stewardship**: culture before antibiotics, avoid pharmacy-bought ciprofloxacin, use narrow-spectrum drugs by sensitivity.
- **GBS bacteriuria**: intrapartum prophylaxis prevents neonatal GBS sepsis.

13. Special Populations

- **First trimester**: no trimethoprim; avoid sulfonamides.
- **Near term (from about 36–38 weeks) and in labour**: no nitrofurantoin; no sulfonamides; use cephalexin.
- **G6PD deficiency** (relatively common in Uganda): avoid nitrofurantoin (maternal and neonatal haemolysis).
- **HIV**: on co-trimoxazole prophylaxis, continue it but do not rely on it as UTI treatment; sterile pyuria → think urinary TB.
- **Sickle cell disease and trait**: higher UTI risk; UTI is a crisis trigger in SCD; urine culture each trimester.
- **Diabetes**: glycosuria; recurrent UTI; check glucose.
- **Penicillin allergy**: nitrofurantoin; cephalexin only if the allergy was mild and non-anaphylactic.
- **Renal impairment**: avoid nitrofurantoin if eGFR below 45; avoid or adjust gentamicin.
- **Breastfeeding**: nitrofurantoin is compatible except for G6PD-deficient or jaundiced/preterm infants.

14. Key Points

High-yield points

- Pregnancy makes bacteriuria dangerous, not commoner: screen and treat ASB.
- A negative dipstick does not exclude ASB: culture.
- Fever plus loin pain = pyelonephritis = admit.
- Culture before the first dose; test of cure 1–2 weeks after.
- Nitrofurantoin for the bladder, not the kidney, and not at term.
- No ciprofloxacin, tetracyclines, first-trimester trimethoprim or term sulfonamides.
- Ceftriaxone IV for pyelonephritis, then oral to complete 10–14 days.
- Sepsis screen and bundle within 1 hour; mRDT for every fever.
- 500 mL boluses with reassessment: ARDS is the hidden killer.
- Prophylaxis after recurrence or pyelonephritis.

Numbers & doses

Item	Value
Significant bacteriuria (MSU)	**10^5 CFU/mL or more** of one organism (lower on catheter specimen; **GBS at any count**)
ASB prevalence	**2–10%** (about **13%** at Mulago)
Untreated ASB → pyelonephritis	**20–40%** (about 1–2% if treated)
Pyelonephritis	**1–2%** of pregnancies; mostly 2nd/3rd trimester; **right** more than left
E. coli share	**70–80%**
Dipstick nitrite sensitivity	Only about **50%**
Microscopy pyuria	**More than 10 pus cells/HPF**; white cell casts = pyelonephritis
Nitrofurantoin	**100 mg MR 12-hourly** (or 50–100 mg 6-hourly standard) × **5–7 days**; avoid from about 36–38 weeks, in labour, G6PD deficiency
Amoxicillin	**500 mg 8-hourly × 7 days** (only if sensitive)
Cephalexin	**500 mg 8–12-hourly × 5–7 days**
Amoxicillin-clavulanate / cefixime	**625 mg 8-hourly** / **200 mg 12-hourly**, 5–7 days
ASB course (WHO)	**7 days**
Pyelonephritis IV	**Ceftriaxone 1 g IV daily (2 g if septic)**; alternative ampicillin 2 g 6-hourly + gentamicin 5 mg/kg daily
IV to oral switch	Afebrile **24–48 h**; total course **10–14 days**
Blood culture positive in pyelonephritis	**15–20%**
Sepsis triggers	T **above 38 or below 36**, HR **above 100**, RR **22 or more**, SBP **100 or below**, SpO_2 **below 95%**, confusion, lactate **above 2** (severe **4 or more**)
Fluid bolus	**500 mL** crystalloid, reassess, up to about **30 mL/kg**
Urine output target	**0.5 mL/kg/h or more**

Item	Value
Creatinine suspicious in pregnancy	Above about **77–80 µmol/L** (normal 35–70)
Review for failure	**48–72 h** (culture, renal ultrasound)
Test of cure	**1–2 weeks** after treatment
Prophylaxis	Nitrofurantoin **50–100 mg** nightly or cephalexin **250–500 mg** nightly until delivery
GBS intrapartum prophylaxis	Benzylpenicillin **3 g IV**, then **1.8 g IV 4-hourly**; clindamycin **900 mg IV 8-hourly** if allergic
Dexamethasone (preterm 24–34 weeks)	**6 mg IM 12-hourly × 4**
Recurrence after pyelonephritis	**6–8%** (higher without prophylaxis)

Exam traps

- **"Ciprofloxacin for UTI."** Wrong in pregnancy: **avoid fluoroquinolones**. Use nitrofurantoin, cephalexin, amoxicillin-clavulanate or ceftriaxone.
- **"Negative dipstick, so no ASB."** Wrong: nitrites are only about 50% sensitive; **culture** is the screen.
- **"Nitrofurantoin for pyelonephritis."** Wrong: poor renal tissue and blood levels. **IV ceftriaxone**.
- **"Nitrofurantoin at 39 weeks."** Wrong: neonatal haemolysis; use cephalexin near term.
- **"Treat pyelonephritis as an outpatient with oral antibiotics."** Wrong: **admit every case**.
- **"The mRDT is positive, so the fever is malaria."** Not necessarily: malaria and pyelonephritis co-exist, and HRP2 tests stay positive after treatment. Examine the loins and dip the urine.
- **"Give 3 L of fluid quickly because she is febrile."** Wrong: pregnant women with pyelonephritis are prone to **ARDS/pulmonary oedema**. **500 mL boluses** with reassessment.
- **"Salbutamol for the contractions."** Wrong: beta-agonists worsen pulmonary oedema. Treat the infection; use **nifedipine** if tocolysis is appropriate.
- **"Single-dose or 3-day treatment for ASB."** Wrong: 5–7 days (WHO 7 days).

Questions seniors ask

On the ward round

1. **"How do you know this is pyelonephritis and not cystitis?"** Fever, rigors, loin pain and renal angle tenderness (upper tract). Cystitis has no fever or loin pain.
2. **"Is she septic?"** State the triggers: temperature, HR, RR, SBP, SpO_2, consciousness, urine output, lactate. One or more with infection = start the sepsis bundle within 1 hour.
3. **"Did you send a urine culture before the first dose?"** It should be sent before antibiotics; it is the only way to catch ESBL and to do a meaningful test of cure.
4. **"What antibiotic, what dose and for how long?"** Ceftriaxone 1 g IV daily (2 g if septic) until afebrile 24–48 hours, then oral cefixime, amoxicillin-clavulanate or cephalexin by sensitivity to complete 10–14 days.
5. **"She is still febrile at 72 hours. What now?"** Check culture (ESBL: meropenem), renal ultrasound for obstruction or abscess, rethink the diagnosis (malaria, chorioamnionitis, appendicitis, pneumonia), urology if obstructed.
6. **"Why is her Hb falling?"** Endotoxin-mediated **haemolysis** (about a quarter of women), plus dilution from IV fluids; also consider malaria.
7. **"What happens after discharge?"** Complete the oral course, **test of cure** in 1–2 weeks, **nightly prophylaxis** until delivery, monthly (or each-trimester) urine culture, look for a cause (diabetes, sickle cell, HIV, stones).

8. **"What does sterile pyuria make you think of?"** Chlamydia/gonorrhoea, recent antibiotics, contamination by vaginal discharge, and **urinary TB** (especially with HIV).

Viva questions

1. **Why is pregnancy a risk factor for pyelonephritis?** Progesterone-induced ureteric dilatation and hypoperistalsis, mechanical compression at the pelvic brim (right more than left), vesicoureteric reflux, glycosuria/aminoaciduria, larger bladder with residual urine.
2. **Why is the right side affected more?** Dextrorotation of the uterus, the sigmoid colon cushions the left ureter, and the right ovarian vein crosses the right ureter.
3. **Define ASB and explain why we treat it in pregnancy.** 10^5 CFU/mL or more of a single organism without symptoms; 20–40% progress to pyelonephritis if untreated; treatment reduces pyelonephritis and low birth weight.
4. **Name antibiotics safe in pregnancy and those to avoid.** Safe: penicillins, cephalosporins, nitrofurantoin (not near term), fosfomycin. Avoid: fluoroquinolones, tetracyclines, trimethoprim (first trimester), sulfonamides (first trimester/term), prolonged aminoglycosides.
5. **Why not nitrofurantoin for pyelonephritis?** It concentrates in urine but achieves poor renal tissue and blood levels.
6. **Why avoid nitrofurantoin at term?** Risk of **haemolytic anaemia in the neonate** (immature glutathione system, G6PD deficiency).
7. **What are the complications of pyelonephritis in pregnancy?** Sepsis, septic shock, ARDS, anaemia, AKI, preterm labour, recurrence.
8. **What do you do if fever persists after 72 hours of IV ceftriaxone?** Review culture/sensitivity (ESBL?), renal ultrasound for obstruction/abscess, reconsider diagnosis (malaria, chorioamnionitis, appendicitis), change antibiotics, urology.
9. **What is a test of cure and when?** Repeat urine culture 1–2 weeks after treatment.
10. **What should be screened for in recurrent UTI?** Diabetes, sickle cell, HIV, stones, anomalies.

Mnemonics

Why pregnancy predisposes: "PROGRESS"

- **P**rogesterone relaxes smooth muscle
- **R**eflux (vesicoureteric)
- **O**bstruction by the gravid uterus (right more than left)
- **G**lycosuria and aminoaciduria
- **R**esidual urine in a larger bladder
- **E**. coli from the nearby perineum
- **S**tasis in dilated ureters
- **S**uppressed immunity (relative)

Antibiotics to avoid: "Stop The Fetal Toxic Antibiotics"

- **S**ulfonamides (first trimester and term)
- **T**rimethoprim (first trimester)
- **F**luoroquinolones
- **T**etracyclines
- **A**minoglycosides (prolonged use)

Safe choices: "Nice Pregnant Cells Face Care": **N**itrofurantoin (not at term), **P**enicillins (amoxicillin, co-amoxiclav), **C**ephalosporins (cephalexin, cefixime, ceftriaxone), **F**osfomycin, **C**linical culture-guided therapy.

Pyelonephritis management: "ADMIT": **A**ll cases admitted; **D**raw cultures (blood, urine) first; **M**edicate IV ceftriaxone; **I**ntake/output (careful fluids, watch for ARDS); **T**est of cure and prophylaxis afterwards.

OSCE tips

- **Urinalysis station**: state the dipstick results systematically (leucocytes, nitrites, protein, blood, glucose, ketones, specific gravity, pH) and interpret each in pregnancy (e.g. protein = UTI or pre-eclampsia; glucose = physiological or diabetes; ketones = vomiting/starvation).
- **Prescribing station**: always state gestation-appropriate drug, dose, route, frequency, duration, test of cure, and safety-net advice.
- **MSU instruction station**: teach the 5 steps of the clean-catch MSU (section 8) and ask her to repeat them back.

15. References

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Why does pregnancy predispose to pyelonephritis?	**Progesterone** relaxes ureteric smooth muscle (dilatation, reduced peristalsis, stasis); the **gravid uterus compresses** the ureters at the pelvic brim; **vesicoureteric reflux**; **glycosuria/aminoaciduria**; larger bladder with residual urine; relative immunosuppression.
2	Why is pyelonephritis in pregnancy more often right-sided?	The uterus is **dextrorotated**, the **sigmoid colon cushions the left ureter**, and the **right ovarian vein** crosses the right ureter, so the right ureter is compressed more.
3	Define asymptomatic bacteriuria.	**10^5 CFU/mL or more of a single uropathogen** on a clean-catch midstream urine culture in a woman **without urinary symptoms**.
4	Why screen and treat ASB in pregnancy but not in most non-pregnant women?	Untreated, **20–40%** of pregnant women with ASB develop **pyelonephritis** (versus about 1–2% if treated); treatment also reduces **low birth weight/preterm birth**.
5	Commonest organism causing UTI in pregnancy?	***Escherichia coli*** (70–80%), then *Klebsiella*, *Proteus*, *Enterobacter*, *Enterococcus*, group B streptococcus, *Staph. saprophyticus*.
6	Recommended duration of treatment for ASB in pregnancy (WHO)?	A **7-day** antibiotic regimen (5–7 days for cystitis), followed by a **test of cure** culture 1–2 weeks later.
7	Nitrofurantoin dose and main cautions in pregnancy?	**100 mg MR orally 12-hourly (or 50–100 mg 6-hourly) for 5–7 days**. **Not for pyelonephritis**; avoid **at term/in labour** and in **G6PD deficiency** (haemolysis).
8	Three oral antibiotics safe for cystitis in pregnancy?	**Nitrofurantoin**, **cephalexin 500 mg 8–12-hourly**, **amoxicillin 500 mg 8-hourly** (if sensitive) or **amoxicillin-clavulanate 625 mg 8-hourly**; also cefixime, fosfomycin 3 g single dose. 5–7 days.
9	Why avoid trimethoprim in the first trimester?	It is a **folate antagonist** (dihydrofolate reductase inhibitor), associated with **neural tube and other congenital defects**.
10	Why avoid sulfonamides near term?	They **displace bilirubin from albumin**, increasing the risk of **neonatal jaundice and kernicterus** (and haemolysis in G6PD deficiency).
11	Why avoid fluoroquinolones and tetracyclines in pregnancy?	**Fluoroquinolones**: cartilage/arthropathy in animal studies. **Tetracyclines**: **tooth discolouration**, impaired fetal bone growth, maternal hepatotoxicity.
12	First-line inpatient treatment of pyelonephritis in pregnancy (Uganda)?	**Admit**; urine and blood cultures, then **IV ceftriaxone 1 g once daily (2 g if septic)** until afebrile 24–48 hours, then oral step-down (cefixime, amoxicillin-clavulanate or cephalexin by sensitivity) to complete **10–14 days**.
13	Why can't nitrofurantoin be used for pyelonephritis?	It concentrates in the **urine** but achieves **poor renal tissue and blood levels**, so it cannot treat parenchymal infection or bacteraemia.
14	Main complications of pyelonephritis in pregnancy?	**Sepsis/septic shock**, **ARDS**/pulmonary oedema, **anaemia** (haemolysis), **AKI**, **preterm labour**, abscess/obstruction, **recurrence**.
15	Why are pregnant women with pyelonephritis prone to ARDS?	**Endotoxin-mediated capillary leak**, plus pregnancy's **low oncotic pressure** and high plasma volume; worsened by **aggressive IV fluids** and **beta-agonist tocolytics**.

#	Question	Answer
16	What do you do if a woman is still febrile after 48–72 hours of IV ceftriaxone?	Check **culture sensitivities** (ESBL?), **renal ultrasound** (obstruction, stone, abscess), reconsider the diagnosis (malaria, chorioamnionitis, appendicitis, pneumonia), change antibiotics (e.g. meropenem per culture), involve urology.
17	Prophylaxis for recurrent UTI in pregnancy?	**Nitrofurantoin 50–100 mg nightly** (switch near term) or **cephalexin 250–500 mg nightly**, until delivery (or up to 6 weeks postpartum), with monthly urine cultures.
18	What should you do with group B streptococcus found in urine?	Treat it now (any colony count) **and** give **intrapartum IV antibiotic prophylaxis** (benzylpenicillin 3 g IV, then 1.8 g 4-hourly until delivery).
19	Internal vs external dysuria?	**Internal** (felt inside the urethra) suggests **UTI**; **external** (urine stinging sore vulval skin) suggests **vulvovaginitis** (candida, trichomonas, herpes).
20	What does sterile pyuria suggest?	White cells without bacterial growth: **chlamydia/gonorrhoea**, **urinary TB** (think HIV), contamination with vaginal discharge, recent antibiotics, stones.
21	Key investigations for suspected pyelonephritis in a Ugandan ANC ward?	Urine dipstick and **culture before antibiotics**, **blood cultures**, FBC, creatinine, **mRDT/blood smear**, glucose, lactate where available, fetal heart/CTG; renal ultrasound if not improving.
22	What are the sepsis-screen triggers in a pregnant woman with suspected pyelonephritis?	Infection plus any of: temperature **above 38 or below 36 °C**, HR **above 100**, RR **22 or more**, SBP **100 mmHg or below**, new confusion, SpO_2 **below 95%**, oliguria (**below 0.5 mL/kg/h**) or lactate **above 2 mmol/L**. Start the **sepsis bundle within 1 hour**.
23	Booking MSU grows 10^5 CFU/mL E. coli; dipstick was nitrite-negative and she has no symptoms. Diagnosis and plan?	**Asymptomatic bacteriuria** (the dipstick is only ~50% sensitive). Treat by sensitivity for **7 days** (e.g. nitrofurantoin 100 mg MR 12-hourly), then a **test-of-cure culture 1–2 weeks later**, and rescreen each trimester.
24	Urine: LE +, nitrites negative, many squamous epithelial cells, mixed growth. What does this mean?	A **contaminated sample**, often from vaginal discharge. Examine for vulvovaginitis, treat the discharge, and **repeat a proper clean-catch MSU** before diagnosing UTI.
25	What urine microscopy finding points to pyelonephritis rather than cystitis?	**White cell casts** (formed in the renal tubules). Pyuria (more than 10 pus cells/HPF) and bacteria occur in both.
26	How should IV fluids be given to a pregnant woman with pyelonephritis and hypotension, and why?	**500 mL crystalloid boluses**, reassessing chest, SpO_2 and BP after each (up to about 30 mL/kg), with a strict fluid balance chart and urine output **0.5 mL/kg/h or more**. Endotoxin capillary leak plus low oncotic pressure make her prone to **pulmonary oedema/ARDS**.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 23-year-old G1 at 14 weeks has a routine booking urine culture growing 10^5 CFU/mL of E. coli sensitive to nitrofurantoin, cephalexin and amoxicillin-clavulanate. She has no urinary symptoms. What is the MOST appropriate management?

- A No treatment because she is asymptomatic
- B Treat with nitrofurantoin (or cephalexin) for 7 days and repeat urine culture 1–2 weeks after treatment
- C Ciprofloxacin 500 mg twice daily for 3 days
- D Single-dose amoxicillin 3 g and no follow-up
- E Admit for IV ceftriaxone

2. A 28-year-old at 22 weeks presents with dysuria, frequency and suprapubic pain for 2 days. Temperature 36.9 °C, no loin tenderness. Dipstick: nitrites positive, leucocytes 2+. Which is the MOST appropriate treatment?

- A Doxycycline 100 mg twice daily for 7 days
- B Ciprofloxacin 500 mg twice daily for 7 days
- C Trimethoprim 200 mg twice daily for 3 days
- D IV ceftriaxone 1 g daily for 14 days
- E Nitrofurantoin 100 mg MR twice daily for 5–7 days after sending an MSU for culture

3. A 25-year-old at 38 weeks has symptomatic cystitis. Her culture grows E. coli sensitive to nitrofurantoin, cephalexin and amoxicillin-clavulanate. Her previous baby needed phototherapy for 'G6PD-related jaundice'. What is the BEST choice? Advanced

- A Nitrofurantoin 100 mg MR twice daily for 7 days
- B Co-trimoxazole 960 mg twice daily for 7 days
- C Cephalexin 500 mg 8-hourly for 7 days
- D Doxycycline 100 mg twice daily for 7 days
- E No treatment until after delivery

4. A 24-year-old at 26 weeks presents with fever 39.2 °C, rigors, right loin pain and vomiting for 2 days. HR 118, BP 104/66, RR 22. Renal angle tenderness on the right. Dipstick: nitrites positive. mRDT negative. What is the MOST appropriate management?

- A Admit; take urine and blood cultures, start IV ceftriaxone, careful IV fluids with fluid balance, paracetamol, and monitor mother and fetus
- B Oral nitrofurantoin and review in 3 days
- C Oral ciprofloxacin as an outpatient
- D IV artesunate only
- E Emergency caesarean section for maternal sepsis

5. A 27-year-old at 30 weeks has been on IV ceftriaxone for pyelonephritis for 72 hours and still spikes fevers to 39 °C with persistent left loin pain. Urine culture grows Klebsiella resistant to ceftriaxone (ESBL). What is the BEST next step? Advanced

A Continue ceftriaxone for another week
B Stop antibiotics and observe
C Add oral nitrofurantoin
D Change to an antibiotic guided by sensitivities (e.g. meropenem) and arrange a renal ultrasound to exclude obstruction or abscess
E Start ciprofloxacin

6. A 26-year-old at 28 weeks is on day 2 of IV ceftriaxone and IV fluids (5 L in 24 hours) for pyelonephritis. She becomes breathless: RR 34, SpO_2 86% on air, bilateral crackles. What is the MOST likely diagnosis? Advanced

A Pulmonary embolism
B Acute respiratory distress syndrome / pulmonary oedema complicating pyelonephritis
C Asthma
D Anxiety
E Amniotic fluid embolism

7. A 22-year-old at 32 weeks with pyelonephritis develops regular painful contractions on day 1 of IV ceftriaxone. Cervix is 2 cm dilated. Temperature 38.8 °C. What is the MOST appropriate management? Advanced

A Salbutamol infusion for tocolysis
B Discharge with oral antibiotics
C Immediate caesarean section
D Stop antibiotics because they cause contractions
E Paracetamol for fever, continue IV antibiotics, give corticosteroids (dexamethasone 6 mg IM 12-hourly × 4), avoid beta-agonists; consider nifedipine only if appropriate

8. A 29-year-old HIV-positive woman at 8 weeks on TLD and co-trimoxazole prophylaxis has symptomatic cystitis. Which antibiotic is MOST appropriate?

A Trimethoprim 200 mg twice daily
B Increase co-trimoxazole to 960 mg twice daily as treatment
C Cephalexin 500 mg 8-hourly for 5–7 days (after sending culture)
D Ciprofloxacin 500 mg twice daily
E Doxycycline 100 mg twice daily

9. A 30-year-old at 24 weeks has had cystitis twice and pyelonephritis once in this pregnancy. She has just completed treatment and the test of cure is negative. What is the MOST appropriate next step?

A Nightly nitrofurantoin 50–100 mg (or cephalexin 250–500 mg) until delivery, monthly urine cultures, and look for an underlying cause (glucose, sickle status, HIV, renal ultrasound)
B No further action
C Weekly ciprofloxacin
D Elective caesarean section at 37 weeks
E Stop all fluids at night to reduce urine output

10. A 21-year-old at 30 weeks presents with fever and right-sided abdominal pain. The HC IV midwife says 'it must be malaria' because the mRDT is positive. Examination shows right renal angle tenderness; urine has nitrites and leucocytes 3+. What is the BEST interpretation?

A Malaria alone; treat with antimalarials only

B Pyelonephritis alone; the mRDT must be false

C Appendicitis; call the surgeons and give no antibiotics

D Possible co-existing malaria and pyelonephritis: treat malaria and admit for IV antibiotics with urine and blood cultures

E Normal pregnancy discomfort

11. A 26-year-old at 34 weeks with pyelonephritis is hypotensive (BP 82/48), HR 132, RR 28, confused, with lactate 4.6 mmol/L after the first litre of fluid. What is the MOST appropriate next step? Advanced

A Continue oral antibiotics and observe on the ward

B Septic shock: call senior/ICU help, oxygen, broad-spectrum IV antibiotics (e.g. ceftriaxone 2 g ± gentamicin), further 500 mL fluid boluses with reassessment, catheter, and vasopressors if hypotension persists

C Immediate caesarean section

D Give 4 L of IV fluid rapidly then reassess

E Start tocolysis

12. A 24-year-old at 20 weeks has urine microscopy showing 30 pus cells per high-power field but repeated cultures show no growth. She has a vaginal discharge and her partner has urethral discharge. What is the MOST likely cause? Advanced

A E. coli pyelonephritis

B Group B streptococcal bacteriuria

C Normal pregnancy finding

D Renal cell carcinoma

E Urethritis/cervicitis from chlamydia or gonorrhoea (sterile pyuria)

13. A 27-year-old at 36 weeks has group B streptococcus (10^4 CFU/mL) on urine culture. She is asymptomatic. What is the MOST appropriate plan?

A No action because the count is below 10^5

B Give intrapartum antibiotics only if she develops fever

C Treat the bacteriuria now and give intrapartum IV benzylpenicillin prophylaxis in labour

D Elective caesarean section

E Treat the baby with antibiotics for 7 days after birth routinely

14. A 23-year-old at 12 weeks with suspected cystitis gives a urine sample. The lab reports mixed growth of three organisms with many squamous epithelial cells. What is the BEST next step?

A Repeat a properly collected clean-catch midstream urine

B Treat all three organisms with broad-spectrum antibiotics

C Diagnose pyelonephritis

D Perform suprapubic aspiration routinely

E Ignore the symptoms

15. A 25-year-old at 30 weeks was treated for pyelonephritis with IV ceftriaxone for 3 days and is now afebrile for 48 hours and eating. What is the MOST appropriate discharge plan?

A Stop antibiotics; she is cured
B Oral ciprofloxacin for 7 days
C Oral nitrofurantoin for 3 days only
D Oral step-down (e.g. cefixime 200 mg 12-hourly) to complete 10–14 days total, test-of-cure culture 1–2 weeks later, then suppressive prophylaxis and regular cultures
E IV ceftriaxone at home for 4 weeks

Answers & explanations

1. B · Treat with nitrofurantoin (or cephalexin) for 7 days and repeat urine culture 1–2 weeks after treatment

B is correct. ASB in pregnancy must be treated (20–40% progress to pyelonephritis if untreated), using a sensitive, pregnancy-safe antibiotic for **7 days** (WHO), then a **test of cure**. **A** ignores the risk. **C**: fluoroquinolones are avoided in pregnancy. **D**: single-dose regimens are less effective, and there is no test of cure. **E**: ASB is treated orally as an outpatient.

2. E · Nitrofurantoin 100 mg MR twice daily for 5–7 days after sending an MSU for culture

E is correct. Uncomplicated **cystitis** is treated orally with a pregnancy-safe agent such as **nitrofurantoin** (fine at 22 weeks), with culture before starting and a test of cure. **A**: tetracyclines stain teeth and affect bone. **C**: trimethoprim is best avoided (folate antagonist; mainly first trimester) and 3 days is short in pregnancy. **D**: IV therapy is for pyelonephritis. **B**: fluoroquinolones are avoided.

3. C · Cephalexin 500 mg 8-hourly for 7 days

C is correct. Cephalexin is safe at term. **A**: nitrofurantoin is avoided **at term** because of **neonatal haemolysis**, especially with a family history suggesting **G6PD deficiency**. **B**: sulfonamides near term risk **kernicterus** and haemolysis. **D**: tetracyclines are contraindicated. **E**: untreated cystitis may progress to pyelonephritis.

4. A · Admit; take urine and blood cultures, start IV ceftriaxone, careful IV fluids with fluid balance, paracetamol, and monitor mother and fetus

A is correct. This is **acute pyelonephritis with sepsis criteria**: admit, cultures first, **IV ceftriaxone** within an hour, careful fluids (ARDS risk), antipyretic and fetal monitoring. **B**: nitrofurantoin does not reach renal tissue; oral outpatient care is unsafe with vomiting and sepsis. **C**: fluoroquinolones are avoided and outpatient care is inappropriate. **D**: the mRDT is negative and there is a clear urinary source. **E**: delivery is not a treatment for pyelonephritis.

5. D · Change to an antibiotic guided by sensitivities (e.g. meropenem) and arrange a renal ultrasound to exclude obstruction or abscess

D is correct. Treatment failure at 48–72 hours needs **culture-guided antibiotics** (ESBL: a carbapenem such as meropenem) and **imaging** for obstruction, stones or abscess (which may need drainage/stenting). **A**: the organism is resistant. **C**: nitrofurantoin does not treat renal parenchyma. **B** is dangerous. **E**: fluoroquinolones are avoided in pregnancy, and ESBL organisms are often co-resistant.

6. B · Acute respiratory distress syndrome / pulmonary oedema complicating pyelonephritis

B is correct. Endotoxin-induced **capillary leak** plus pregnancy's low oncotic pressure and **large-volume IV fluids** cause **ARDS/pulmonary oedema** (1–2% of pyelonephritis). Stop fluids, oxygen, senior/ICU review, consider diuretic if overload. **A**: PE causes breathlessness but usually with a clear chest. **C**: no history, and wheeze rather than crackles. **D** would not cause hypoxia. **E** occurs in labour/delivery with collapse and coagulopathy.

7. E · Paracetamol for fever, continue IV antibiotics, give corticosteroids (dexamethasone 6 mg IM 12-hourly × 4), avoid beta-agonists; consider nifedipine only if appropriate

E is correct. Fever and inflammation trigger contractions, which often settle once fever is controlled. **Steroids** below 34 weeks improve neonatal outcome. **A**: beta-agonists increase the risk of **pulmonary oedema/ARDS** in pyelonephritis. **C**: no obstetric indication. **D** is wrong. **B**: she needs inpatient care.

8. C · Cephalexin 500 mg 8-hourly for 5–7 days (after sending culture)

C is correct. Cephalexin is safe in the first trimester and does not rely on the antifolate class to which her bacteria may already be exposed. **A**: trimethoprim is a folate antagonist to avoid in the **first trimester**. **B**: treatment-dose co-trimoxazole in the first trimester is best avoided, and bacteria in women on prophylaxis are often resistant. **D** and **E** are contraindicated in pregnancy.

9. A · Nightly nitrofurantoin 50–100 mg (or cephalexin 250–500 mg) until delivery, monthly urine cultures, and look for an underlying cause (glucose, sickle status, HIV, renal ultrasound)

A is correct. Recurrent UTI in pregnancy warrants **suppressive prophylaxis**, **monthly cultures** and a search for **risk factors** (diabetes, sickle cell, HIV, stones, anomalies). **B** leaves a high recurrence risk. **C**: fluoroquinolones are avoided. **D**: UTI is not an indication for CS. **E**: fluid restriction increases stasis and risk.

10. D · Possible co-existing malaria and pyelonephritis: treat malaria and admit for IV antibiotics with urine and blood cultures

D is correct. In Uganda **malaria and pyelonephritis can co-exist**, and an mRDT can remain positive after recent treated malaria; clear urinary findings with loin tenderness mean **pyelonephritis must be treated too**. **A** misses a sepsis risk. **B**: a positive mRDT should not be dismissed. **C**: appendicitis is a differential for right-sided pain, but urinary findings and renal angle tenderness favour pyelonephritis; antibiotics should not be withheld. **E** is unsafe.

11. B · Septic shock: call senior/ICU help, oxygen, broad-spectrum IV antibiotics (e.g. ceftriaxone 2 g ± gentamicin), further 500 mL fluid boluses with reassessment, catheter, and vasopressors if hypotension persists

B is correct. Hypotension despite fluids and lactate above 4 mean **septic shock**: follow the sepsis bundle with senior/ICU input and **vasopressors** if needed; left lateral tilt. **A** is inadequate. **C**: an unstable mother should be resuscitated first; delivery is for obstetric indications. **D**: large uncontrolled fluid volumes risk **ARDS**; give boluses with reassessment. **E** is irrelevant and harmful.

12. E · Urethritis/cervicitis from chlamydia or gonorrhoea (sterile pyuria)

E is correct. Sterile pyuria with discharge in her and her partner suggests an **STI** (chlamydia/gonorrhoea); manage with syndromic treatment safe in pregnancy (e.g. ceftriaxone + azithromycin) and partner treatment. **A**: would grow on culture and cause fever/loin pain. **C**: significant pyuria is not normal. **D** is very unlikely. **B**: GBS grows on culture.

13. C · Treat the bacteriuria now and give intrapartum IV benzylpenicillin prophylaxis in labour

C is correct. GBS bacteriuria at any count indicates heavy colonisation: treat now and give **intrapartum antibiotic prophylaxis** (benzylpenicillin 3 g IV then 1.8 g 4-hourly) to prevent early-onset neonatal GBS sepsis. **A**: the 10^5 threshold does not apply to GBS. **B** is too late for prevention. **D**: CS does not replace prophylaxis and is not indicated. **E**: routine neonatal antibiotics are not recommended if the mother received adequate prophylaxis.

14. A · Repeat a properly collected clean-catch midstream urine

A is correct. Mixed growth with epithelial cells indicates **contamination**; teach clean-catch technique (separate labia, clean front to back, midstream sample, lab within 2 hours) and repeat. If symptomatic, empirical treatment can start after the repeat sample is sent. **B** treats contamination. **C**: no clinical features of pyelonephritis. **D**: invasive and unnecessary. **E**: symptoms still need assessment.

15. D · Oral step-down (e.g. cefixime 200 mg 12-hourly) to complete 10–14 days total, test-of-cure culture 1–2 weeks later, then suppressive prophylaxis and regular cultures

D is correct. Complete **10–14 days** with an oral agent to which the organism is sensitive, then **test of cure**, **prophylaxis** (recurrence 6–8%) and regular cultures. **A**: 3 days is inadequate for pyelonephritis. **C**: nitrofurantoin does not treat renal parenchyma and 3 days is too short. **B**: fluoroquinolones are avoided. **E**: unnecessary and impractical.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Asymptomatic bacteriuria at booking

Harriet, 20, G1 at 13 weeks, books at a HC IV in Mbale. She is well. The booking urine dipstick shows nitrites positive and leucocytes 1+. The hospital lab 30 km away can do cultures.

1. What is the likely diagnosis, and why does it matter in pregnancy?

> **Asymptomatic bacteriuria** (confirm by culture: 10^5 CFU/mL or more of one organism). It matters because pregnancy's ureteric dilatation and stasis mean **20–40% progress to pyelonephritis** if untreated, with risks of sepsis and preterm birth; treatment reduces pyelonephritis and low birth weight.

2. What history and checks do you add?

> Symptoms (internal dysuria, frequency beyond her usual, fever, loin pain), previous UTIs, diabetes risk factors, **sickle status**, HIV status, drug allergies, G6PD history, vaginal discharge. Check BP, temperature, urine protein and glucose. Teach **clean-catch MSU** technique and send to the lab within 2 hours (or refrigerate).

3. Culture grows E. coli 10^5 CFU/mL, resistant to amoxicillin and co-trimoxazole, sensitive to nitrofurantoin and cephalexin. What do you prescribe?

> **Nitrofurantoin 100 mg MR orally 12-hourly for 7 days** (or 50–100 mg 6-hourly), or **cephalexin 500 mg 8-hourly for 7 days**. Not amoxicillin (resistant), not co-trimoxazole or trimethoprim (resistant; first trimester), never ciprofloxacin or doxycycline. Advise fluids, regular voiding, and return if fever or loin pain.

4. What follow-up does she need?

> **Test-of-cure culture 1–2 weeks** after finishing, then culture **each trimester** (or monthly if recurrent). If ASB recurs, re-treat by sensitivities and consider **nightly prophylaxis** until delivery. Look for causes if recurrent (glucose, sickle, HIV, renal ultrasound).

Take-away: Screen at booking, confirm with culture, treat ASB for 7 days with a pregnancy-safe drug chosen by sensitivity, and always do a test of cure.

Case 2: Pyelonephritis with sepsis at 26 weeks

Night admission at Mengo: Proscovia, 24, G2P1 at 26 weeks, with 2 days of fever, rigors, right loin pain and vomiting. T 39.2 °C, HR 118, RR 22, BP 104/66, SpO_2 97%. Right renal angle tenderness. Uterus soft, no contractions, FHR 170.

1. Summarise the likely diagnosis and immediate differentials.

> **Acute right pyelonephritis with sepsis** (MEOWS triggers). Differentials: **malaria** (can co-exist), appendicitis (right-sided, higher in pregnancy), chorioamnionitis (uterus non-tender), pneumonia, cholecystitis, renal stone with infected obstruction.

2. What do you do in the first hour?

> Senior informed; **IV access**; bloods: FBC, creatinine, electrolytes, **blood cultures**, **mRDT**, glucose, lactate; **urine for culture before antibiotics**; **IV ceftriaxone 2 g** within the hour; **Ringer's lactate 500 mL bolus** and reassess; **paracetamol 1 g**; antiemetic; catheter or strict intake/output; oxygen if SpO_2 falls; left lateral position; FHR monitoring.

3. What do you monitor over the next 48 hours, and which complications do you look for?

> MEOWS 1–4-hourly, **SpO_2 and RR** (ARDS/pulmonary oedema, especially with fluids), urine output (AKI), temperature curve, Hb (**haemolytic anaemia**), creatinine, **contractions** (preterm labour; steroids if delivery likely, avoid salbutamol), fetal heart (tachycardia should settle as fever falls).

4. On day 3 she is still febrile at 38.9 °C. Culture shows E. coli resistant to ceftriaxone (ESBL). What now?

> Switch to a **culture-guided agent** (e.g. **meropenem 1 g IV 8-hourly**), request **renal ultrasound** (obstruction, stone, abscess; hydronephrosis out of proportion to gestation), involve urology if obstructed (stent/nephrostomy), recheck malaria and other sources.

5. She improves and is afebrile for 48 hours. How do you discharge her?

> Oral step-down according to sensitivities to complete **10–14 days** (for an ESBL organism, options are limited: nitrofurantoin is not adequate for pyelonephritis; discuss with microbiology, e.g. complete IV therapy or use an oral agent to which it is sensitive and which is safe in pregnancy). **Test of cure** 1–2 weeks later, **nightly prophylaxis** (e.g. nitrofurantoin 50–100 mg if sensitive) until delivery, **monthly cultures**, safety-net advice.

Take-away: Pyelonephritis in pregnancy is a sepsis emergency: cultures first, IV antibiotics within the hour, careful fluids, watch for ARDS and preterm labour, and escalate early if not better at 48–72 hours.

Case 3: Recurrent UTIs and an underlying cause

Sylvia, 30, G4P3 at 28 weeks, attends ANC at a regional referral hospital with her third symptomatic UTI of this pregnancy. She is overweight and her last baby weighed 4.2 kg. She has never had an HIV test in this pregnancy.

1. What underlying causes should you look for, and why?

> **Diabetes/GDM** (glycosuria feeds bacteria; previous macrosomia and obesity are risk factors), **HIV** (immunosuppression), **sickle cell disease/trait** (papillary damage, concentrating defect), **urinary stones or anomalies**, incomplete bladder emptying, sexual/hygiene factors, and inadequate treatment (resistant organism, no test of cure).

2. Which investigations do you order?

> MSU culture and sensitivity; **75 g OGTT** (she is 28 weeks with risk factors); **HIV test**; **Hb electrophoresis/sickling test** if status unknown; creatinine; **renal ultrasound** (stones, hydronephrosis out of proportion, anomalies).

3. Culture grows Proteus sensitive to cephalexin; OGTT confirms GDM; renal ultrasound shows a right renal pelvis stone. How do you manage her?

> Treat the UTI: **cephalexin 500 mg 8-hourly for 7 days**; test of cure. Start **prophylaxis**: **cephalexin 250–500 mg nightly** until delivery. Manage **GDM** (diet, monitoring, metformin/insulin as needed). **Stone**: urology review; Proteus (urease-producing) favours struvite stones; stent/nephrostomy only if obstruction or infection fails to settle; definitive treatment after delivery.

4. What follow-up is needed after delivery?

> Continue prophylaxis up to 6 weeks postpartum if advised; **postpartum GDM testing at 6–13 weeks**; **renal imaging** after physiological dilatation resolves and stone management; counsel about early urine screening in future pregnancies.

Take-away: Recurrent UTI in pregnancy is a clue: treat and prevent (nightly prophylaxis), then hunt for the cause (diabetes, HIV, sickle cell, stones, anomalies).

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.